A QUASI-COMPACT SURJECTIVE FLAT MORPHISM IS SUBMERSIVE ON THE UNDERLYING TOPOLOGICAL SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 4843-4853 and proof span 4855-4878 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The counted statement contains a reference environment holding the author's two citations (SGA1 and EGA); the project's own shared preamble defines that environment as a comment environment, so no bibliography is attached and none is needed.

Editorial note. The statement's 'resp. closed' clause and the proof's use of complements are the author's wording and are preserved.

Editorial note. 4 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The question of openness and closedness of a subset of the target being local on the target, and quasi-compactness of the source over an affine base. Used at proof lines 3072-3079. Permits replacing the base by an affine open and covering the quasi-compact source by finitely many affine opens, including after replacing the source by their disjoint union.
- Generalizations lift along flat morphisms, cited from the same chapter, and the topological lemma that images of sets stable under generalization are the complements of sets stable under specialization. Cited at proof lines 3090-3093. Shows that the complement of the candidate closed subset is stable under generalization, hence the subset itself is stable under specialization.
- The algebra lemma that a subset of an affine scheme stable under specialization is closed. Cited at proof lines 3094-3095. Closes the affine case of the argument, from which the general case follows by the initial reductions.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 00HY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 660–660. Same-chapter reference used in the retained author text at source line(s) 4877. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 03HV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4581–4581. Same-chapter reference used in the retained author text at source line(s) 1315, 4469, 4803, 4871, 14824. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0065: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 233–233. Same-chapter reference used in the retained author text at source line(s) 4872. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0062: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 230–230. Same-chapter reference used in the retained author text at source line(s) 4875. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to Y$ be a quasi-compact, surjective, flat morphism. A subset $T \subset Y$ is open (resp. closed) if and only $f^{-1}(T)$ is open (resp. closed). In other words, f is a submersive morphism.*

Proof. The question is local on Y, hence we may assume that Y is affine. In this case X is quasi-compact as f is quasi-compact. Write $X = X_1 \cup \ldots \cup X_n$ as a finite union of affine opens. Then $f' : X' = X_1 \amalg \ldots \amalg X_n \to Y$ is a surjective flat morphism of affine schemes. Note that for $T \subset Y$ we have $(f')^{-1}(T) = f^{-1}(T) \cap X_1 \amalg \ldots \amalg f^{-1}(T) \cap X_n$. Hence, $f^{-1}(T)$ is open if and only if $(f')^{-1}(T)$ is open. Thus we may assume both X and Y are affine.

Let $f : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$ be a surjective morphism of affine schemes corresponding to a flat ring map $A \to B$. Suppose that $f^{-1}(T)$ is closed, say $f^{-1}(T) = V(J)$ for $J \subset B$ an ideal. Then $T = f(f^{-1}(T)) = f(V(J))$ is the image of $\mathrm{Spec}(B/J) \to \mathrm{Spec}(A)$ (here we use that f is surjective). On the other hand, generalizations lift along f (Lemma 03HV). Hence by Topology, Lemma 0065 we see that $Y \setminus T = f(X \setminus f^{-1}(T))$ is stable under generalization. Hence T is stable under specialization (Topology, Lemma 0062). Thus T is closed by Algebra, Lemma 00HY. □